

Home

Shorts

Subscriptions

- Velocipecti
- Breaking Italy
- Roman Paolucci
- Lucy
- MIT OpenCourseW...
- Dimitri Bianco
- Mr. RIP
- Show more

You

- Your channel
- History
- Playlists
- Watch Later
- Liked videos
- Your videos
- Downloads
- Show more

Explore

- Music
- Films
- Hype
- Show more

More from YouTube

Skip navigation

YouTube Premium

YouTube Music

YouTube Kids

Report history

About Press Copyright

Contact us Creator

Advertise Developers

Terms Privacy

Policy & Safety

How YouTube works

Test new features

© 2026 Google LLC YouTube, a

Google company



Dynamical Systems

(with Machine Learning)

by Steve Brunton

Playlist • 24 videos • 247,949 views

Modeling complex systems from data is an age old pursuit. Machine learning is rapidly improving our abilities to model complex systems.

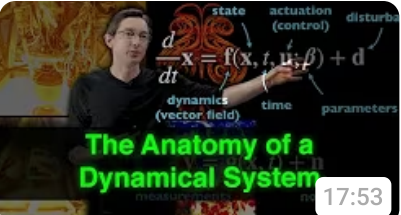
Play all



Data-Driven Dynamical Systems Overview

Steve Brunton

116k 8 yr ago



The Anatomy of a Dynamical System

Steve Brunton

122k 5 yr ago



Simulating the Lorenz System in Matlab

Steve Brunton

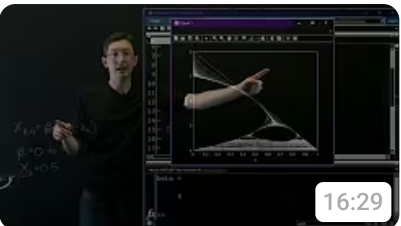
52k 8 yr ago



Discrete-Time Dynamical Systems

Steve Brunton

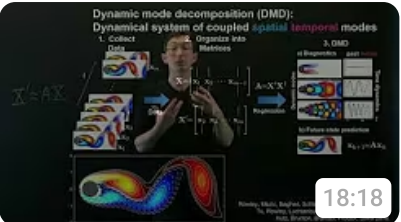
43k 8 yr ago



Simulating the Logistic Map in Matlab

Steve Brunton

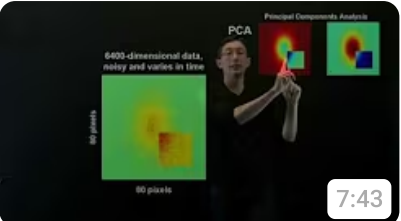
41k 8 yr ago



Dynamic Mode Decomposition (Overview)

Steve Brunton

115k 8 yr ago



Dynamic Mode Decomposition (Examples)

Steve Brunton

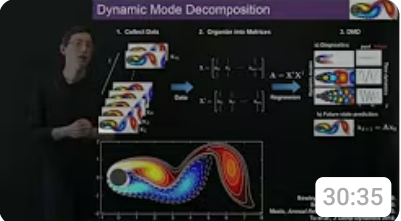
38k 8 yr ago



Dynamic Mode Decomposition (Code)

Steve Brunton

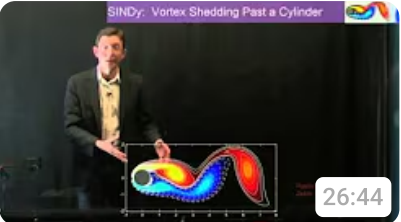
36k 8 yr ago



Compressed Sensing and Dynamic Mode Decomposition

Steve Brunton

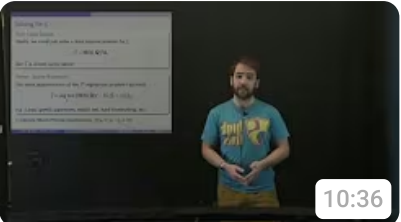
44k 10 yr ago



Sparse Identification of Nonlinear Dynamics (SINDy)

Steve Brunton

85k 10 yr ago



PDE FIND

Nathan Kutz

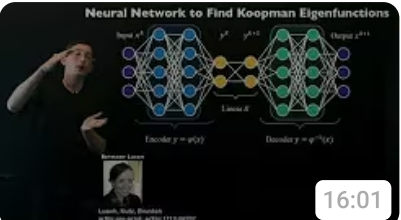
8.7k 9 yr ago



Koopman Spectral Analysis (Overview)

Steve Brunton

55k 8 yr ago



Koopman Spectral Analysis (Representations)

Steve Brunton

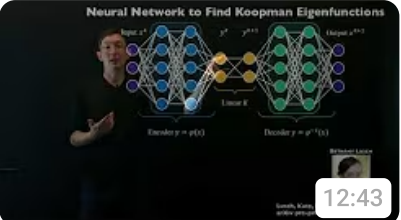
24k 8 yr ago



Koopman Spectral Analysis (Control)

Steve Brunton

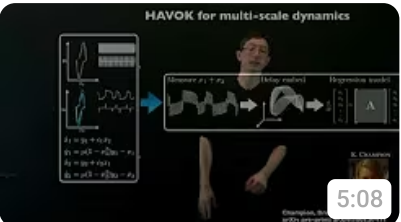
13k 8 yr ago



Koopman Spectral Analysis (Continuous Spectrum)

Steve Brunton

12k 8 yr ago



Koopman Spectral Analysis (Multiscale systems)

Steve Brunton

9.1k 8 yr ago

Home

Shorts

Subscriptions

- Velocipliedi
- Breaking Italy
- Roman Paolucci
- Lucy
- MIT OpenCourseW...
- Dimitri Bianco
- Mr. RIP
- Show more

You

- Your channel
- History
- Playlists
- Watch Later
- Liked videos
- Your videos
- Downloads
- Show more

Explore

- Music
- Films
- Hype
- Show more

More from YouTube

- YouTube Premium
- YouTube Music
- YouTube Kids

Report history

About

Press

Copyright

Contact us

Creator

Advertise

Developers

Terms

Privacy

Policy & Safety

How YouTube works

Test new features

© 2026 Google LLC YouTube, a Google company



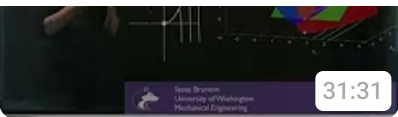
Dynamical Systems (with Machine Learning)

by Steve Brunton

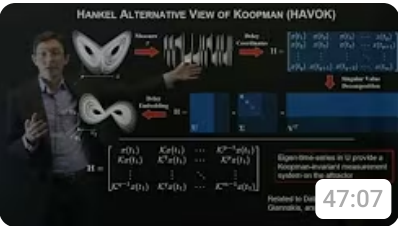
Playlist • 24 videos • 247,949 views

Modeling complex systems from data is an age old pursuit. Machine learning is rapidly improving our ability

Play all



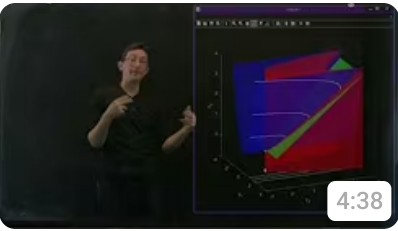
▷ 56k 10 yr ago



Hankel Alternative View of Koopman (HAVOK) Analysis [FULL]

Steve Brunton

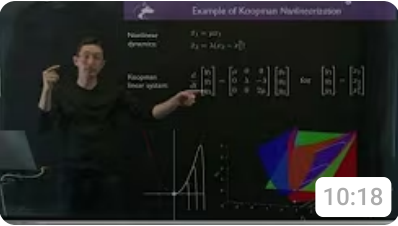
▷ 31k 10 yr ago



Koopman Observable Subspaces & Nonlinearization

Steve Brunton

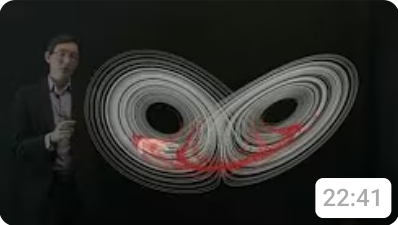
▷ 10k 10 yr ago



Koopman Operator Optimal Control

Steve Brunton

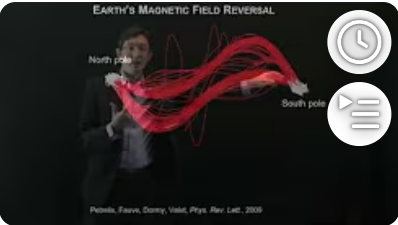
▷ 24k 10 yr ago



Hankel Alternative View of Koopman (HAVOK) Analysis [SHORT]

Steve Brunton

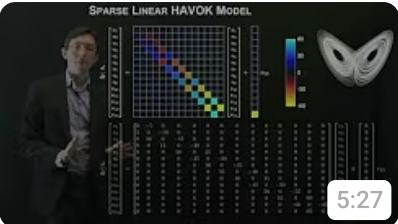
▷ 6.5k 10 yr ago



Magnetic field reversal and Measles outbreaks: HAVOK models of chaos

Steve Brunton

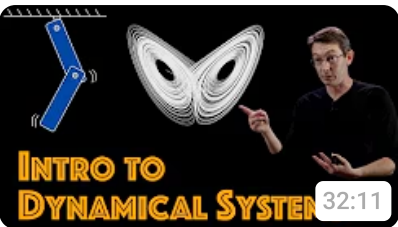
▷ 6.5k 10 yr ago



Linear model for chaotic Lorenz system [HAVOK]

Steve Brunton

▷ 5.8k 10 yr ago



Topics in Dynamical Systems: Fixed Points, Linearization, Invariant Manifolds, Bifurcations & Chaos

Steve Brunton

▷ 41k 3 yr ago